

Slide 1: Title Slide

CSE400 - Fundamentals of Probability in Computing

Lecture 21 and L22: Inequalities and Tail Bound

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Slide 2: Acknowledgement

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Reference Book:

- *Introduction to Probability for Computing*
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Slide 3: Outline

1. Tail and Motivation of Tail Bound
2. Markov's Inequality
 - Definition and Proof
 - Example
3. Chebyshev's Inequality
 - Definition and Proof
 - Example
4. Chernoff Bound and Poisson Tail
 - Definition and Proof
 - Example
5. Jensen's Inequality
 - Convex Functions
 - Proof and Example
6. Case Study and System-Level Understanding

Slide 4: Tails – Definition

Definition: The tail of a random variable X is:

$$P(X > x)$$

Slide 5: Tails – Motivation

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Slide 6: Tails – Key Idea

Key idea:

- Mean $\rightarrow$ average behavior (easy)
- Tail $\rightarrow$ rare extreme events (hard)

Slide 7: Tails – Difficulty

Why are tails hard to compute?

- Requires summing many small probabilities

Slide 8–10: Tails Example 1 (Binomial Case)

Question: Suppose you are distributing n jobs among n servers at random. What is the probability that a particular server gets $\geq k$ jobs?

Intuition:

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is a **tail event (rare overload)**

Model:

$$X \sim \text{Binomial}(n, p)$$

Exact Tail Probability:

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Slide 11–13: Tails Example 2 (Poisson Case)

Question: Jobs arrive according to a Poisson process with rate λ . What is the probability that $\geq k$ jobs arrive in one hour?

Intuition:

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes bursts occur

This is a **tail event (sudden spike in load)**

Model:

$$X \sim \text{Poisson}(\lambda)$$

Exact Tail Probability:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Slide 14: Why Tail Bounds?

Exact tail probabilities:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Observation:

- These are complicated, long sums

Slide 15: Idea of Tail Bounds

Instead of computing exact value:

- Find an **upper bound**
- Easier to compute
- Still gives a guarantee

Goal:

$$\text{Exact: } P(X \geq k) = ?$$

$$\text{Bound: } P(X \geq k) \leq \text{simple expression}$$

Key Idea: Approximate safely instead of computing exactly

Slide 16–17: Running Example Setup

We will evaluate different bounds on the same example.

Experiment:

- Flip a fair coin n times

Distribution:

X = number of heads

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean:

$$E[X] = \frac{n}{2}$$

Question:

$$P\left(X \geq \frac{3n}{4}\right)$$

Slide 18: Markov's Inequality – Definition

Key Idea:

- Uses only the mean
- Large values are rare

Markov Inequality:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Slide 19: Markov's Inequality – Proof

Consider:

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

Restrict to $x \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

Replace $x \geq a$ with a :

$$\begin{aligned} &\geq \sum_{x=a}^{\infty} a \cdot p_X(x) \\ &= a \sum_{x=a}^{\infty} p_X(x) \\ &= a \cdot P(X \geq a) \end{aligned}$$

Thus:

$$\begin{aligned} E[X] &\geq a \cdot P(X \geq a) \\ P(X \geq a) &\leq \frac{E[X]}{a} \end{aligned}$$

Slide 20–21: Markov on Running Example

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad E[X] = \frac{n}{2}$$

We want:

$$P\left(X \geq \frac{3n}{4}\right)$$

Apply Markov:

$$\begin{aligned} P\left(X \geq \frac{3n}{4}\right) &\leq \frac{E[X]}{3n/4} \\ &= \frac{n/2}{3n/4} \\ &= \frac{2}{3} \end{aligned}$$

Observation: This is a poor bound (does not depend on n)

Slide 22–24: Markov Real-Life Example

Scenario: Income distribution

$$E[X] = 10,000$$

Question: Fraction earning $\geq 50,000$

Apply Markov:

$$\begin{aligned} P(X \geq 50,000) &\leq \frac{10,000}{50,000} \\ &= 0.2 \end{aligned}$$

Result: At most 20% of people earn $\geq 50,000$

Slide 25–29: Markov Solved Example 1

Problem: Average grade = 70, A-grade cutoff = 90

Step 1

Let $X \geq 0$, $E[X] = 70$

Step 2

Markov inequality:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 3

Substitute:

$$P(X \geq 90) \leq \frac{70}{90}$$

Step 4

Simplify:

$$= \frac{7}{9}$$

Result

$$P(X \geq 90) \leq 0.778$$

Slide 30–34: Markov Solved Example 2

Problem: Average requests = 5000/sec Find $P(X \geq 20,000)$

Step 1

$$E[X] = 5000$$

Step 2

$$a = 20,000$$

Step 3

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 4

$$\begin{aligned} P(X \geq 20,000) &\leq \frac{5000}{20,000} \\ &= \frac{1}{4} \end{aligned}$$

Result

$$P(X \geq 20,000) \leq 0.25$$

Slide 35: Markov Homework Problem

Given: $E[X] = 40,000$

(a) Bound $P(X \geq 120,000)$

(b) Can Markov bound $P(X < 10,000)$?

Hint: Markov applies to right tail $P(X \geq a)$

Slide 36–38: Chebyshev's Inequality

Key Idea:

- Uses mean + variance
- Controls deviation from mean

Chebyshev Inequality:

$$P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Proof Idea:

$$\begin{aligned} P(|X - \mu| \geq a) &= P((X - \mu)^2 \geq a^2) \\ &\leq \frac{E[(X - \mu)^2]}{a^2} \\ &= \frac{\text{Var}(X)}{a^2} \end{aligned}$$

Slide 39: Chebyshev on Running Example

$$\begin{aligned} X &\sim \text{Binomial}(n, \tfrac{1}{2}) \\ \mu &= \frac{n}{2}, \quad \text{Var}(X) = \frac{n}{4} \\ P\left(X \geq \frac{3n}{4}\right) &= P\left(X - \frac{n}{2} \geq \frac{n}{4}\right) \\ &\leq P\left(|X - \mu| \geq \frac{n}{4}\right) \\ &\leq \frac{n/4}{(n/4)^2} = \frac{1}{n} \end{aligned}$$

Slide 40–42: Chebyshev Real-Life Example

Given:

$$\mu = 70, \quad \sigma = 10$$

$$P(X \leq 40 \text{ or } X \geq 100) = P(|X - 70| \geq 30)$$

$$\leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$$

Result: At most $\frac{1}{9}$ students are far from mean

Slide 43–47: Chebyshev Solved Example 1

Given:

$$\mu = 70, \quad \sigma = 5 \Rightarrow \sigma^2 = 25$$

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

$$\leq \frac{25}{400} = \frac{1}{16}$$

Result

$$P(X \geq 90) \leq 0.0625$$

Slide 48–52: Chebyshev Solved Example 2

Given:

$$\mu = 50, \quad \sigma^2 = 16$$

$$X \notin [30, 70] \Rightarrow |X - 50| \geq 20$$

$$\leq \frac{16}{400} = \frac{1}{25}$$

Result

$$P(|X - 50| \geq 20) \leq 0.04$$

Slide 53: Chebyshev Homework

Given:

$$\mu = 100, \quad \sigma = 2$$

Defective if:

$$|X - 100| > 6$$

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Slide 54–58: Chernoff Bound – Key Idea

- Markov $\rightarrow$ uses X
- Chebyshev $\rightarrow$ uses $(X - \mu)^2$
- Chernoff $\rightarrow$ uses e^{tX}

Insight:

- Exponential grows very fast
- Large values amplified
- Tail events amplified

Better transformation tighter bound

Slide 59–61: Chernoff Core Trick

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Slide 62–64: Chernoff Bound (Final Form)

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight:

- Holds for all $t > 0$
- Choose best $t \rightarrow$ tightest bound

Much stronger than Markov & Chebyshev

Slide 65–68: Chernoff Lower Tail

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Slide 69–70: Chernoff for Poisson Tail

$$X \sim \text{Poisson}(\lambda)$$

$$\begin{aligned} E[e^{tX}] &= \sum_{i=0}^{\infty} e^{ti} \cdot \frac{e^{-\lambda} \lambda^i}{i!} \\ &= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!} \\ &= e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda(e^t - 1)} \end{aligned}$$